

ALISTAIR BAILEY

An engineer by training, I have since worked primarily as an informatician and research scientist. I am currently a Research Fellow in Microfluidic Hydrogen-Deuterium Exchange at the University of Southampton.¹

The topic of my research career has been the role of HLA antigen processing and presentation in disease recognition by T cells. In cancer² this has focused on HLA-presented tumour antigens³, and in infectious disease the focus has been HLA-presented viral⁴ and bacterial antigens. Exploiting these targets has the potential for enhancing personalised therapies, vaccine development and understanding allergy.

I have contributed to research into COVID19⁵, skin sensitization to chemical allergens⁶, asthma⁷ and contagious cancer in the Tasmanian Devil⁸.

My core skills are processing and analysing data from whole exome sequencing, RNAseq, scRNAseq and proteomics assays. My workflow combines command line tools with micromamba, the R programming language and git version control.

Proteomics data I have curated, deposited and I am the data controller for is deposited at the PRoteomics IDentifications Archive⁹. Whole Exome and RNAseq data I have curated, deposited and I am the data controller for is deposited at the European Genome-phenome Archive¹⁰.

I am a Data and Software Carpentry¹¹ instructor and I have also created and delivered my own workshops to teach foundational R coding and data science skills¹² to bioscientists and web design¹³ to librarians.



View this CV online with links at ab604.uk/cv/cv.html

CONTACT

✉ alistair@ab604.uk

🌐 ab604.uk

📞 0000-0003-0023-8679

🐙 github.com/ab604

☁️ [@ab604.uk](https://twitter.com/ab604)

🎓 EDUCATION

- 2017 • **Carpentries Instructor**
Worldwide 📍 The Carpentries
- I trained as a Carpentries¹⁴ instructor as part of their volunteer led mission to increase global capacity in essential data and computational skills for conducting efficient, open, and reproducible research.
- 2016 • **Machine Learning**
Stanford University 📍 Coursera
- 10 week online introduction to machine learning.
- 2015 • **Data Science Specialization**
John Hopkins University 📍 Coursera
- 12 month online set of courses on data science using R, git and command line tools.
- 2013
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2008 • **PhD, Immunology**
Cancer Sciences, University of Southampton 📍 Southampton, UK
- Thesis: Relating the structure, function and dynamics of the MHC Class I antigen presenting molecule.
- 2008
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2005 • **BEng, Civil Engineering**
University of Southampton 📍 Southampton, UK
- First Class Honours in Civil Engineering.

LANGUAGE SKILLS

R	
Markdown	
Bash	
Latex	
Git	
HTML	
CSS	
Python	
JavaScript	
SQL	

Made with the R package [pagedown](#).

The source code is available on github.com/ab604/abailey-cv.

The fonts are Inter and Permanent Marker

Last updated on 2025-12-29.

2005
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2004

Engineering, Science & Mathematics Foundation Year

University of Southampton

📍 Southampton, UK

- Maths and physics foundation year preparation for undergraduate study.

1994
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1992

BTEC ND Audio-Visual Production

Bournemouth & Poole College of Art & Design

📍 Bournemouth, UK

- Foundation course in film, photography, TV and radio production.



RESEARCH EXPERIENCE

2025

Research Fellow

Cancer Sciences, University of Southampton

📍 Southampton, UK

- Research Fellow in Microfluidic HDX

2023

Research Fellow

School of Biological Sciences, University of Southampton

📍 Southampton, UK

- scRNAseq analysis of T-cell response to neutrophil exposure.
Bioinformatician maternity leave cover for Medical Research Council funded project.

2022
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2018

Research Fellow

Centre for Proteomic Research/Cancer Sciences,
University of Southampton

📍 Southampton, UK

- Cancer Research UK Accelerator: this project aims to identify potential treatment targets for hard to treat cancers using multi-omics methods. In this project our focus was on oesophageal, lung and neuroendocrine cancers.

As an informatician I processed, analysed and managed data from whole exome sequencing, RNAseq, scRNAseq and proteomics.

For sequencing fastq data, my workflow comprised of a mixture of command line tools using bash scripts and R/RStudio. I followed the Broad Institute Best Practices for genomic data analysis¹⁵ and Cornell Bioinformatics Core¹⁶. For proteomics data, my workflow used Peaks Studio¹⁷, and post-process in R and RStudio.

Scripts and processed data were managed using git version control. Raw data was deposited along with processed outputs in PRoteomics IDentifications Archive¹⁸ and the European Phenome-Genome Archive¹⁹.

We also developed our method to identify treatment targets for infectious diseases from influenza and bacterial proteins. In 2020 I also worked to develop a COVID19 test using proteomics methods.

2018
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2015

Research Fellow
Centre for Proteomic Research/Cancer Sciences,
University of Southampton

📍 Southampton, UK

- Developed peptidomics methodology at the UoS for research into the role of MHC molecules in skin sensitisation to chemical allergy.

2015
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2013

Research Fellow
Cancer Sciences, University of Southampton

📍 Southampton, UK

- MRC Centenary Fellow

TEACHING + EDUCATION EXPERIENCE

2024
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2023

Learning Technologist
University of Southampton

📍 Southampton, UK

- Part-time role supporting Engagement Librarians in the design, development and deployment of webpages and other electronic resources.

2024

Webpage Design²⁰
University of Southampton

📍 Southampton, UK

- I created a webpage design workshop and materials for Librarians at the University of Southampton

2022
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2021

BIOL 2013: Introduction to bioinformatics
University of Southampton

📍 Southampton, UK

- I taught the undergraduate introduction to bioinformatics module on variant discovery using the University Galaxy Server.

2020
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2019

Coding Together²¹
University of Southampton

📍 Southampton, UK

- I created and taught an eight week series of collaborative workshops to teach foundational R coding and data science skills based on Carpentries materials.

2019
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2018

Academic Support Tutor
IntoUniversity Millbrook

📍 Southampton, UK

- IntoUniversity²² supports young people from disadvantaged backgrounds to attain either a university place or another chosen aspiration. I volunteered as an academic support tutor for secondary school learners.

2018

Software Carpentry
Umeå University

📍 Umeå, Sweden

- Taught R for Reproducible Research and assisted in Command Line Basics.

I enjoy teaching foundational coding and data science skills to researchers and developing evidence-based best practices. I am especially interested in helping novices and making coding more accessible to all.

- 2018

British Society for Proteomics 2018 Data Science Workshop²³

University of Bradford 📍 Bradford, UK

 - I created and taught a proteomics data science workshop including introduction to R, Volcano plots, heatmaps and peptide logos.

- 2017

Data Carpentry

University of Southampton 📍 Southampton, UK

 - Taught R for Reproducible Research and assisted in Command Line Basics and git.

- 2017

Data Carpentry

University of Southampton 📍 Southampton, UK

 - Taught R for Reproducible Research and assisted in introduction to SQL.

- 2017

Software Carpentry

University of Southampton 📍 Southampton, UK

 - Assisted with python and git for reproducible research.



RESEARCH DATA

- Immunopeptidomic analysis of influenza A virus infected human tissues identifies internal proteins as a rich source of HLA ligands²⁴, Publicly released**

 - Proteomics data: PRIDE Project PXD022884²⁵

- Identification of neoantigens in esophageal adenocarcinoma²⁶, Publicly released**

 - Proteomics data: PRIDE Project ID PXD031108²⁷
 - WES & RNAseq data EGA Study ID EGAS00001005957

- Characterization of the Class I MHC Peptidome Resulting From DNCB Exposure of HaCaT Cells²⁸, Publicly released**

 - Proteomics data: PRIDE Project PXD021373²⁹

- Neoantigen identification in pancreatic neuroendocrine tumours, Unreleased pending publication**

 - Proteomics data: PRIDE Project ID PXD037449
 - WES & RNAseq data EGA Study ID EGAS00001006722

- **Immunopeptidomics guided identification of neoantigens in non-small cell lung cancer, Unreleased pending publication**
 - Proteomics data: PRIDE Project ID PXD028990
 - WES & RNAseq data EGA Study ID EGAS00001005499
- **Immunopeptidomics of a brain tumour cell line to identify HLA presented Zika, Unreleased pending publication**
 - Proteomics data: PRIDE Project ID PXD037627
- **Non-small cell lung cancer global proteomics, Unreleased pending publication**
 - Proteomics data: PRIDE Project ID PXD054390
- **Oesophageal adenocarcinoma global proteomes, Unreleased pending publication**
 - Proteomics data: PRIDE Project ID PXD054428

INDUSTRY EXPERIENCE

- 2012 ● **Internship**
Microsoft Research  Cambridge, UK
 - Helped develop computational model of MHC I peptide selection.
- 2012 | 2004 ● **Freelance Satellite Communications Engineer**
Globecast  London, UK
 - I continued to work as an engineer in broadcast TV from 2004 and 2012 on major events such as the Olympics and Football World Cup.
- 2004 | 2000 ● **Satellite Communications Engineer**
Globecast  London, UK
 - Full time engineer working in global broadcast TV primarily on sports, news and live entertainment events.
- 2000 | 1995 ● **Film and television post-production engineer**
Telecine  London, UK
 - I trained as an engineer to operate various TV & film post-production equipment.

I have worked in a variety of roles ranging from engineering to research scientist. I like collaborative environments where I can learn from my peers.



PUBLICATIONS

- 2025 • **Molecular Mechanisms of Disinfectant Resistance in *Klebsiella pneumoniae*³⁰**
JAC-Antimicrobial Resistance
- Daniel Noel, Alistair Bailey, Benjamin Nicholas, Paul Skipp, Bill Keevil, Sandra Wilks
- 2025 • **Identification of natural Zika virus peptides presented on the surface of paediatric brain tumour cells by HLA class I³¹**
PLOS One
- Matt Sherwood, José Ramos-Castañeda, Ben Nicholas, Alistair Bailey, Thiago Giove Mitsugi, Carolini Kaid, Oswaldo K. Okamoto, Paul Skipp, Rob M. Ewing
- 2025 • **Droplet microfluidic hydrogen/deuterium exchange for investigating protein dynamics with millisecond precision³²**
ChemRxiv
- Dietmar Hammerschmid, Alistair Bailey, Jakub Sys, Simon I.R. Lane, Max Saito, Theo Hornsey, Niall Hanrahan, Andy van Hateren, Howard Broughton, Alfonso Espada, Eamonn Reading, Jonathan West
- 2025 • **Evidence of focusing the MHC class I immunopeptidome by tapasin³³**
Frontiers in Immunology
- Rachel Darley, Patricia T. Illing, Patrick Duriez, Alistair Bailey, Anthony W. Purcell, Andy van Hateren, Tim Elliott.
- 2025 • **Comparative analysis of protein expression between oesophageal adenocarcinoma and normal adjacent tissue³⁴**
PLOS One
- Ben Nicholas, Alistair Bailey, Katy J. McCann, Robert C. Walker, Peter Johnson, Tim Elliott, Tim J. Underwood, Paul Skipp
- 2025 • **Comparative analysis of transcriptomic and proteomic expression between two non-small cell lung cancer subtypes³⁵**
Journal of Proteome Research
- Ben Nicholas, Alistair Bailey, Katy J McCann, Peter Johnson, Tim Elliott, Christian Ottensmeier and Paul Skipp
- 2024 • **Proteogenomics guided identification of functional neoantigens in non-small cell lung cancer³⁶**
bioRxiv
- Ben Nicholas, Alistair Bailey, Katy J McCann, Oliver Wood, Eve Currall, Peter Johnson, Tim Elliott, Christian Ottensmeier, Paul Skipp

- 2022 ● **Operation Moonshot: rapid translation of a SARS-CoV-2 targeted peptide immunoaffinity liquid chromatography-tandem mass spectrometry test from research into routine clinical use³⁷**
Clinical Chemistry and Laboratory Medicine
- Jenny Hällqvist, Benjamin I. Nicholas, Alistair Bailey et al.
- 2022 ● **Identification of neoantigens in esophageal adenocarcinoma³⁸**
Immunology
- Ben Nicholas, Alistair Bailey, Katy J. McCann, Oliver Wood, Robert C. Walker, Robert Parker, Nicola Ternette, Tim Elliott, Tim J. Underwood, Peter Johnson, Paul Skipp
- 2022 ● **Analysis of cell-specific peripheral blood biomarkers in severe allergic asthma identifies innate immune dysfunction³⁹**
Clinical & Experimental Allergy
- Ben Nicholas, Jane Guo, Hyun-Hee Lee, Alistair Bailey, Rene de Waal Malefyt, Milenko Cicmil, Ratko Djukanovic
- 2022 ● **Immunopeptidomic analysis of influenza A virus infected human tissues identifies internal proteins as a rich source of HLA ligands⁴⁰**
PLoS Pathogens
- Ben Nicholas, Alistair Bailey, Karl J. Staples, Tom Wilkinson, Tim Elliott, Paul Skipp.
- 2021 ● **The differentiation state of the Schwann cell progenitor drives phenotypic variation between two contagious cancers⁴¹**
PLOS Pathogens
- Rachel S. Owen, Sri H. Ramarathinam, Alistair Bailey, Annalisa Gastaldello, Kathryn Hussey, Paul J. Skipp, Anthony W. Purcell, Hannah V. Siddle
- 2021 ● **Characterization of the Class I MHC Peptidome Resulting From DNCB Exposure of HaCaT Cells⁴²**
Toxicological Sciences
- Alistair Bailey, Ben Nicholas, Rachel Darley, Erika Parkinson, Ying Teo, Maja Aleksic, Gavin Maxwell, Tim Elliott, Michael Arden-Jones, Paul Skipp.
- 2021 ● **The immunopeptidomes of two transmissible cancers and their host have a common, dominant peptide motif⁴³**
Immunology
- Annalisa Gastaldello, Sri H. Ramarathinam, Alistair Bailey, Rachel Owen, Steven Turner, N. Kontouli, Tim Elliott, Paul Skipp, Anthony W. Purcell, Hannah V. Siddle.

- 2019 ● **Dynamically Driven Allostery in MHC Proteins: Peptide-Dependent Tuning of Class I MHC Global Flexibility⁴⁴**
Frontiers in Immunology
- Cory M. Ayres, Esam T. Abualrous, Alistair Bailey, Christian Abraham, Lance M. Hellman, Steven A. Corcelli, Frank Noé, Tim Elliott, Brian M. Baker.
- 2017 ● **Direct evidence for conformational dynamics in major histocompatibility complex class I molecules⁴⁵**
JBC
- Andy van Hateren, Malcolm Anderson, Alistair Bailey, Jörn M. Werner, Paul Skipp, Tim Elliott.
- 2017 ● **Recent advances in Major Histocompatibility Complex class I antigen presentation: Plastic MHC molecules and TAPBPR mediated quality control⁴⁶**
F1000 Research
- Andy van Hateren, Alistair Bailey, Tim Elliott.
- 2015 ● **Selector function of MHC I molecules is determined by protein plasticity⁴⁷**
Scientific Reports
- Alistair Bailey, Neil Dalchau, Rachel Carter, Stephen Emmott, Andrew Phillips, Jörn M. Werner, Tim Elliott
- 2014 ● **Two Polymorphisms Facilitate Differences in Plasticity between Two Chicken Major Histocompatibility Complex Class I Proteins⁴⁸**
PLoS One
- Alistair Bailey, Andy van Hateren, Tim Elliott, Jörn M. Werner.
- 2013 ● **A Mechanistic Basis for the Co-evolution of Chicken Tapasin and Major Histocompatibility Complex Class I Proteins⁴⁹**
JBC
- Andy van Hateren, Rachel Carter, Alistair Bailey, Nasia Kontouli, Anthony P. Williams, Jim Kaufman, Tim Elliott.
- 2010 ● **The cell biology of major histocompatibility complex class I assembly: towards a molecular understanding⁵⁰**
Tissue Antigens
- A. Van Hateren, E. James, A. Bailey, A. Phillips, N. Dalchau, T. Elliott



1. <https://www.soton.ac.uk>
2. <https://www.cancerresearchuk.org/funding-for-researchers/accelerator-award/portfolio-funded-projects-outputs>

3. <https://doi.org/10.1111/imm.13578>
4. <https://doi.org/10.1371/journal.ppat.1009894>
5. <https://doi.org/10.1515/cclm-2022-1000>
6. <https://doi.org/10.1093/toxsci/kfaa184>
7. <https://doi.org/10.1111/cea.14197>
8. <https://doi.org/10.1111/imm.13307>
9. <https://www.ebi.ac.uk/pride/>
10. <https://ega-archive.org/>
11. <https://carpentries.org/>
12. <https://ab604.github.io/docs/coding-together-2019/>
13. <https://ab604.github.io/webpage-design/>
14. <https://carpentries.org/>
15. <https://gatk.broadinstitute.org/hc/en-us>
16. <https://abc.med.cornell.edu/>
17. <https://www.bioinfor.com/peaks-studio/>
18. <https://www.ebi.ac.uk/pride/>
19. <https://ega-archive.org/>
20. <https://ab604.github.io/webpage-design/>
21. <https://ab604.github.io/docs/coding-together-2019/>
22. <https://intouniversity.org/>
23. https://ab604.github.io/docs/bspr_workshop_2018/index.html
24. <https://doi.org/10.1371/journal.ppat.1009894>
25. <https://www.ebi.ac.uk/pride/archive/projects/PXD022884>
26. <https://doi.org/10.1111/imm.13578>
27. <https://www.ebi.ac.uk/pride/archive/projects/PXD031108>
28. <https://doi.org/10.1093/toxsci/kfaa184>
29. <https://www.ebi.ac.uk/pride/archive/projects/PXD021373>
30. <https://doi.org/10.1093/jacamr/dlaf247>
31. <https://doi.org/10.1371/journal.pone.0335726>
32. <https://doi.org/10.26434/chemrxiv-2025-j9k30>
33. <https://doi.org/10.3389/fimmu.2025.1563789>
34. <https://doi.org/10.1371/journal.pone.0318572>
35. <https://doi.org/10.1021/acs.jproteome.4c00773>
36. <https://doi.org/10.1101/2024.05.30.596609>
37. <https://doi.org/10.1515/cclm-2022-1000>
38. <https://doi.org/10.1111/imm.13578>
39. <https://doi.org/10.1111/cea.14197>
40. <https://doi.org/10.1371/journal.ppat.1009894>
41. <https://journals.plos.org/plospathogens/article?id=10.1371/journal.ppat.1010033>
42. <https://doi.org/10.1093/toxsci/kfaa184>
43. <https://doi.org/10.1111/imm.13307>
44. <https://doi.org/10.3389/fimmu.2019.00966>
45. <https://doi.org/10.1074/jbc.M117.809624>
46. <https://doi.org/10.12688/f1000research.10474.1>
47. <https://doi.org/10.1038/srep14928>
48. <https://doi.org/10.1371/journal.pone.0089657>
49. <https://doi.org/10.1074/jbc.M113.474031>
50. <https://doi.org/10.1111/j.1399-0039.2010.01550.x>